

MAVEN PRACTICAL 2

Practical # 2: Maven + Docker Integration

```
Command Prompt
C:\Users\aryan>mkdir maven_home1
C:\Users\aryan>cd maven_home1
C:\Users\aryan\maven_home1>mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: C:\Users\aryan\maven_home1
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: docker-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\aryan\maven_home1\docker-demo
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.252 s
[INFO] Finished at: 2026-03-31T12:17:38+05:30
[INFO]
C:\Users\aryan\maven_home1>dir
Volume in drive C is Windows-SSD
Volume Serial Number is E4F7-83BD

Directory of C:\Users\aryan\maven_home1

31-03-2026 12:17 <DIR> .
```

```
Command Prompt
31-03-2026 12:17 <DIR> .
31-03-2026 12:17 <DIR> ..
31-03-2026 12:17 <DIR> docker-demo
0 File(s) 0 bytes
3 Dir(s) 119,592,980,480 bytes free

C:\Users\aryan\maven_home1>cd docker-demo
C:\Users\aryan\maven_home1\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO] -----< com.example:docker-demo >-----
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ docker-demo ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ docker-demo ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ docker-demo ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ docker-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO]
[INFO] T E S T S
[INFO]
```

```
Command Prompt
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.016 s -- in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.5.0:jar (default-jar) @ docker-demo ---
[INFO] Building jar: C:\Users\aryan\maven_home1\docker-demo\target\docker-demo-1.0-SNAPSHOT.jar
[INFO] BUILD SUCCESS
[INFO] Total time: 2.664 s
[INFO] Finished at: 2026-03-31T12:21:03+05:30
```

```
C:\Users\aryan\maven_home1\docker-demo>cd src/main/java/com/example

C:\Users\aryan\maven_home1\docker-demo\src\main\java\com\example>dir
Volume in drive C is Windows-SSD
Volume Serial Number is E4F7-83BD

Directory of C:\Users\aryan\maven_home1\docker-demo\src\main\java\com\example

31-03-2026  12:17    <DIR>          .
31-03-2026  12:17    <DIR>          ..
31-03-2026  12:17                187 App.java
               1 File(s)              187 bytes
               2 Dir(s)  119,899,365,376 bytes free

C:\Users\aryan\maven_home1\docker-demo\src\main\java\com\example>notepad App.java
```

```
Command Prompt
C:\Users\aryan\maven_home1\docker-demo\src\main\java\com\example>cd ../../../../..

C:\Users\aryan\maven_home1\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO] -----< com.example:docker-demo >-----
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ docker-demo ---
[INFO] Deleting C:\Users\aryan\maven_home1\docker-demo\target
[INFO] --- resources:3.4.0:resources (default-resources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ docker-demo ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO] --- resources:3.4.0:testResources (default-testResources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ docker-demo ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO] --- surefire:3.5.4:test (default-test) @ docker-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] T E S T S
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s -- in com.example.AppTest
[INFO] Results:
```

```
Command Prompt
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ docker-demo ---
[INFO] Building jar: C:\Users\aryan\maven_home1\docker-demo\target\docker-demo-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.445 s
[INFO] Finished at: 2026-03-31T12:24:10+05:30
[INFO]
C:\Users\aryan\maven_home1\docker-demo>notepad Dockerfile
C:\Users\aryan\maven_home1\docker-demo>ls
'ls' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\aryan\maven_home1\docker-demo>dir
Volume in drive C is Windows-SSD
Volume Serial Number is E4F7-83BD

Directory of C:\Users\aryan\maven_home1\docker-demo

31-03-2026 12:24 <DIR> .
31-03-2026 12:17 <DIR> ..
31-03-2026 12:24      129 Dockerfile.txt
31-03-2026 12:17      664 pom.xml
31-03-2026 12:17 <DIR> src
31-03-2026 12:24 <DIR> target
                2 File(s)      793 bytes
                4 Dir(s)  119,899,688,960 bytes free

C:\Users\aryan\maven_home1\docker-demo>ren Dockerfile.txt Dockerfile
C:\Users\aryan\maven_home1\docker-demo>notepad pom.xml
C:\Users\aryan\maven_home1\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:docker-demo >-----
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
```

```
Command Prompt
C:\Users\aryan\maven_home1\docker-demo>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:docker-demo >-----
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ docker-demo ---
[INFO] Deleting C:\Users\aryan\maven_home1\docker-demo\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ docker-demo ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ docker-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\aryan\maven_home1\docker-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ docker-demo ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ docker-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.028 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```

Command Prompt
[INFO] --- jar:3.3.0:jar (default-jar) @ docker-demo ---
[INFO] Building jar: C:\Users\aryan\maven_home1\docker-demo\target\docker-demo-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- dockerfile:1.4.13:build (default) @ docker-demo ---
[WARNING] 1 problem was encountered while building the effective model for org.javassist:javassist:jar:3.18.1-GA during dependency collection step for plugin (use -X to see details)
[INFO] dockerfile: null
[INFO] contextDirectory: C:\Users\aryan\maven_home1\docker-demo
[INFO] Building Docker context C:\Users\aryan\maven_home1\docker-demo
[INFO] Path(dockerfile): null
[INFO] Path(contextDirectory): C:\Users\aryan\maven_home1\docker-demo
[INFO]
[INFO] Image will be built as docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Step 1/4 : FROM eclipse-temurin:21-jdk
[INFO]
[INFO] Pulling from library/eclipse-temurin
[INFO] Digest: sha256:3b0a98dfbdf1067c20a7854cec159551777d2ee1381bc76cd4bd0719f543b148
[INFO] Status: Image is up to date for eclipse-temurin:21-jdk
[INFO] --> 3b0a98dfbdf1
[INFO] Step 2/4 : WORKDIR /app
[INFO]
[INFO] --> Using cache
[INFO] --> 9d45f671b0bf
[INFO] Step 3/4 : COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar
[INFO]
[INFO] --> 4bc695e8f69b
[INFO] Step 4/4 : ENTRYPOINT ["java","-jar","app.jar"]
[INFO]
[INFO] --> Running in leab207bcea7
[INFO] --> Removed intermediate container leab207bcea7
[INFO] --> 1fbde1f4a260
[INFO] Successfully built 1fbde1f4a260
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Detected build of image with id 1fbde1f4a260
[INFO] Building jar: C:\Users\aryan\maven_home1\docker-demo\target\docker-demo-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 16.128 s
[INFO] Finished at: 2026-03-31T12:28:24+05:30

```

```

Command Prompt
C:\Users\aryan\maven_home1\docker-demo>dir target
Volume in drive C is Windows-SSD
Volume Serial Number is E4F7-B3BD

Directory of C:\Users\aryan\maven_home1\docker-demo\target

31-03-2026 12:28 <DIR> .
31-03-2026 12:28 <DIR> ..
31-03-2026 12:28 <DIR> classes
31-03-2026 12:28 <DIR> docker
31-03-2026 12:28 2,916 docker-demo-1.0-SNAPSHOT-docker-info.jar
31-03-2026 12:28 2,453 docker-demo-1.0-SNAPSHOT.jar
31-03-2026 12:28 <DIR> generated-sources
31-03-2026 12:28 <DIR> generated-test-sources
31-03-2026 12:28 <DIR> maven-archiver
31-03-2026 12:28 <DIR> maven-status
31-03-2026 12:28 <DIR> surefire-reports
31-03-2026 12:28 <DIR> test-classes
31-03-2026 12:28 2 File(s) 5,369 bytes
31-03-2026 12:28 10 Dir(s) 119,897,337,856 bytes free

C:\Users\aryan\maven_home1\docker-demo>mvn dockerfile:build
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/com/spotify/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/com/spotify/maven-metadata.xml (511 B at 450 B/s)
[INFO]
[INFO] -----< com.example:docker-demo >-----
[INFO] Building docker-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- dockerfile:1.4.13:build (default-cli) @ docker-demo ---
[WARNING] 1 problem was encountered while building the effective model for org.javassist:javassist:jar:3.18.1-GA during dependency collection step for plugin (use -X to see details)
[INFO] dockerfile: null
[INFO] contextDirectory: C:\Users\aryan\maven_home1\docker-demo
[INFO] Building Docker context C:\Users\aryan\maven_home1\docker-demo
[INFO] Path(dockerfile): null
[INFO] Path(contextDirectory): C:\Users\aryan\maven_home1\docker-demo
[INFO]
[INFO] Image will be built as docker-demo:1.0-SNAPSHOT
[INFO]
[INFO] Step 1/4 : FROM eclipse-temurin:21-jdk

```

```
Command Prompt
[INFO] Pulling from library/eclipse-temurin
[INFO] Digest: sha256:3b0a98dfbdf1067c20a7854cec159551777d2ee1381bc76cd4bd0719f543b148
[INFO] Status: Image is up to date for eclipse-temurin:21-jdk
[INFO] --> 3b0a98dfbdf1
[INFO] Step 2/4 : WORKDIR /app
[INFO] --> Using cache
[INFO] --> 9d45f671b0bf
[INFO] Step 3/4 : COPY target/docker-demo-1.0-SNAPSHOT.jar app.jar
[INFO] --> Using cache
[INFO] --> 4bc695e8f69b
[INFO] Step 4/4 : ENTRYPOINT ["java","-jar","app.jar"]
[INFO] --> Using cache
[INFO] --> 1fbde1f4a260
[INFO] Successfully built 1fbde1f4a260
[INFO] Successfully tagged docker-demo:1.0-SNAPSHOT
[INFO] Detected build of image with id 1fbde1f4a260
[INFO] Successfully built docker-demo:1.0-SNAPSHOT
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.165 s
[INFO] Finished at: 2026-03-31T12:29:10+05:30
[INFO] -----

C:\Users\aryan\maven_home1\docker-demo>docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
aryan0z/apl:v1.1                    cd1dba651b30                     119MB       31.7MB       U
aryan123/php_4677:latest            ad185c7d14bc                     707MB       176MB
compose1-web:latest                 84758797e894                     175MB       45.2MB
docker-demo:1.0-SNAPSHOT            1fbde1f4a260                     675MB       211MB
eclipse-temurin:21-jdk              3b0a98dfbdf1                     679MB       214MB
httpd:latest                        96b1e8f69ee3                     177MB       47.6MB       U
mongo:6                             03cda579c8ca                     1.06GB      273MB       U
my-image:latest                     239e0ef2ae36                     230MB       73.6MB
my-mysql:latest                     433817d8cfe4                     1.08GB      233MB
mysql:5.7                           4bc6bc963e6d                     700MB       149MB       U
nginx:latest                        c881927c4077                     240MB       65.7MB
node-mongo-compose-app:latest       bcd13e013e61                     1.63GB      406MB       U
```

```
C:\Users\aryan\maven_home1\docker-demo>docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
aryan0z/apl:v1.1                    cd1dba651b30                     119MB       31.7MB       U
aryan123/php_4677:latest            ad185c7d14bc                     707MB       176MB
compose1-web:latest                 84758797e894                     175MB       45.2MB
docker-demo:1.0-SNAPSHOT            1fbde1f4a260                     675MB       211MB
eclipse-temurin:21-jdk              3b0a98dfbdf1                     679MB       214MB
httpd:latest                        96b1e8f69ee3                     177MB       47.6MB       U
mongo:6                             03cda579c8ca                     1.06GB      273MB       U
my-image:latest                     239e0ef2ae36                     230MB       73.6MB
my-mysql:latest                     433817d8cfe4                     1.08GB      233MB
mysql:5.7                           4bc6bc963e6d                     700MB       149MB       U
nginx:latest                        c881927c4077                     240MB       65.7MB
node-mongo-compose-app:latest       bcd13e013e61                     1.63GB      406MB       U
php_4677:latest                     eef0835ab1a4                     707MB       176MB       U
ubl:latest                          7adf2f3c9817                     412MB       115MB
ubuntu:latest                       cd1dba651b30                     119MB       31.7MB       U
wordpress:latest                   ffe0dca1f0f                      1.08GB      275MB       U

C:\Users\aryan\maven_home1\docker-demo>docker run docker-demo:1.0-SNAPSHOT
Hello from Docker + Maven 0Y3f

C:\Users\aryan\maven_home1\docker-demo>
```